

Supplementary Material for “When Does Sparsification Help Community Detection? Separating Genuine Gains from Degree-Mechanical Artifacts”

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1 Omitted Proofs

Notation follows the main text: $s(e) = 1/d_u + 1/d_v$; $E_{\text{intra}}, E_{\text{inter}}$ with n_1, n_2 , $m = n_1 + n_2$; mean scores $\mu_{\text{intra}}, \mu_{\text{inter}}$, global mean $\bar{s}$; separation $\delta = \mu_{\text{intra}} - \mu_{\text{inter}} > 0$; retention probabilities $p(e) = \alpha s(e)/\bar{s}$ in the unclipped regime. Theorem numbers refer to the main text.

1.1 Theorem 1, parts (b) and (c)

Proof. Part (c): Let $Y = |E' \cap E_{\text{inter}}| = \sum_{e \in E_{\text{inter}}} X_e$ where $X_e = \mathbf{1}[e \in E']$ are independent Bernoulli random variables. Then

$$\text{Var}(Y) = \sum_{e \in E_{\text{inter}}} p(e)(1-p(e)) \leq \sum_{e \in E_{\text{inter}}} p(e) = \frac{\alpha n_2 \mu_{\text{inter}}}{\bar{s}}, \quad (1)$$

and Chebyshev's inequality gives

$$\mathbb{P}[|Y - \mathbb{E}[Y]| > n_2 \epsilon] \leq \frac{\text{Var}(Y)}{n_2^2 \epsilon^2} \leq \frac{\alpha \mu_{\text{inter}}}{n_2 \bar{s} \epsilon^2}. \quad (2)$$

The intra-community bound is identical.

Part (b): Define empirical preservation rates $\hat{p}_{\text{inter}} = Y/n_2$ and $\hat{p}_{\text{intra}} = |E' \cap E_{\text{intra}}|/n_1$. By part (c), for any $\epsilon > 0$,

$$\mathbb{P}[|\hat{p}_{\text{inter}} - \mathbb{E}[\hat{p}_{\text{inter}}]| > \epsilon] \leq \frac{\alpha \mu_{\text{inter}}}{n_2 \bar{s} \epsilon^2} \rightarrow 0 \quad (3)$$

as $n_2 \rightarrow \infty$, and similarly for $\hat{p}_{\text{intra}}$. Hence $\hat{p}_{\text{inter}} = \frac{\alpha \mu_{\text{inter}}^{(n)}}{\bar{s}^{(n)}}(1 + o_P(1))$ and $\hat{p}_{\text{intra}} = \frac{\alpha \mu_{\text{intra}}^{(n)}}{\bar{s}^{(n)}}(1 + o_P(1))$; note that $\bar{s}^{(n)}$, a convex combination of $\mu_{\text{intra}}^{(n)}$ and $\mu_{\text{inter}}^{(n)}$, is bounded in $[\mu_{\min}, \mu_{\max}]$, and it cancels in the ratio.

To apply the continuous mapping theorem to the ratio, we verify that $\hat{p}_{\text{intra}}$ is bounded away from zero with high probability. Since $\mu_{\text{intra}} \geq \mu_{\min} > 0$ and $\bar{s} \leq 2$ (as $s(e) \leq 2$ for any edge), $\mathbb{E}[\hat{p}_{\text{intra}}] \geq \alpha \mu_{\min}/2 > 0$. Taking $\epsilon = \alpha \mu_{\min}/4$, the concentration bound gives

$$\mathbb{P}\left[\hat{p}_{\text{intra}} < \frac{\alpha \mu_{\min}}{4}\right] \leq \mathbb{P}\left[|\hat{p}_{\text{intra}} - \mathbb{E}[\hat{p}_{\text{intra}}]| > \frac{\alpha \mu_{\min}}{4}\right] \rightarrow 0. \quad (4)$$

The continuous mapping theorem, applied to $g(x, y) = x/y$ on $\{(x, y) : y > 0\}$, then yields

$$\text{Ratio}_n = \frac{\hat{p}_{\text{inter}}}{\hat{p}_{\text{intra}}} \xrightarrow{P} \frac{\mu_{\text{inter}}^*}{\mu_{\text{intra}}^*} < 1. \quad (5)$$

1.2 Theorem 2 (full derivation of part (a))

Proof. Under unit weights, $F^{(1)} = n_1/m = n_1/(n_1 + n_2)$. Under DSpar weighting with $w_e = s(e)/\bar{s}$, the normalization $\bar{s}$ cancels between numerator and denominator:

$$F^{(s)} = \frac{\sum_{e \in E_{\text{intra}}} s(e)}{\sum_{e \in E} s(e)} = \frac{n_1 \mu_{\text{intra}}}{n_1 \mu_{\text{intra}} + n_2 \mu_{\text{inter}}}. \quad (6)$$

The inequality $F^{(s)} > F^{(1)}$ is equivalent, by cross-multiplication with all terms positive, to

$$n_1 \mu_{\text{intra}}(n_1 + n_2) > n_1(n_1 \mu_{\text{intra}} + n_2 \mu_{\text{inter}}) \quad (7)$$

$$n_1 n_2 \mu_{\text{intra}} > n_1 n_2 \mu_{\text{inter}}, \quad (8)$$

which holds by DSpar separation. The explicit formula follows by direct subtraction:

$$F^{(s)} - F^{(1)} = \frac{n_1 \mu_{\text{intra}}}{n_1 \mu_{\text{intra}} + n_2 \mu_{\text{inter}}} - \frac{n_1}{n_1 + n_2} \quad (9)$$

$$= n_1 \cdot \frac{\mu_{\text{intra}}(n_1 + n_2) - (n_1 \mu_{\text{intra}} + n_2 \mu_{\text{inter}})}{(n_1 \mu_{\text{intra}} + n_2 \mu_{\text{inter}})(n_1 + n_2)} \quad (10)$$

$$= \frac{n_1 n_2 (\mu_{\text{intra}} - \mu_{\text{inter}})}{m(n_1 \mu_{\text{intra}} + n_2 \mu_{\text{inter}})}. \quad (11)$$

Parts (b) and (c) are immediate from $Q = F - G$. $\square$

1.3 Theorem 3

Proof. Let $X_{\text{intra}} = |E' \cap E_{\text{intra}}|$ and $X_{\text{inter}} = |E' \cap E_{\text{inter}}|$. By linearity of expectation (Theorem 1(a)),

$$\mathbb{E}[X_{\text{intra}}] = \frac{\alpha n_1 \mu_{\text{intra}}}{\bar{s}}, \quad \mathbb{E}[X_{\text{inter}}] = \frac{\alpha n_2 \mu_{\text{inter}}}{\bar{s}}. \quad (12)$$

By the concentration bound of Theorem 1(c), both X_{intra}/n_1 and X_{inter}/n_2 converge in probability to their expectations. Writing

$$F' = \frac{X_{\text{intra}}/n_1}{X_{\text{intra}}/n_1 + (n_2/n_1)(X_{\text{inter}}/n_2)}, \quad (13)$$

part (a) follows from the continuous mapping theorem, with X_{intra}/n_1 bounded away from zero in probability exactly as in the proof of Theorem 1(b); the target quantity in part (a) is centered at finite- n values, so no convergence of n_2/n_1 is required.

For part (b), the assumed convergence $n_2/n_1 \rightarrow \rho$ gives

$$F'_n \xrightarrow{P} \frac{\mu_{\text{intra}}^*}{\mu_{\text{intra}}^* + \rho \mu_{\text{inter}}^*}. \quad (14)$$

Since $F_n^{(1)} = 1/(1 + n_2/n_1) \rightarrow 1/(1 + \rho)$ and $\mu_{\text{intra}}^* > \mu_{\text{inter}}^*$, the claimed inequality follows. $\square$